

DISCOVER UNSPOKEN HABITS

Users need not state a habit explicitly

Interaction Stream

Structured tool events
Parsed natural-language events

Sliding Event Counter

(object, verb) counts
recent summaries
windowed evidence

Habit Discovery

Frequency Filter

domain-aware
trigger verbs
count $\geq n_{\min}$

Deterministic Generalization

same-extension
e.g., edit *.py

Schema-checked LLM Discovery

typed habits
+ variants
domain fallbacks

Typed Habit Store

trigger - context
consequence
confidence - frequency
recency
workflow / temporal
emotional / associative
variants - lifecycle status

feedback to habit state

CONTROL PROACTIVE INTERVENTION

Structured decision with explicit suppression paths

habit + pending state

current event

Feedback Cooldown
global; checked first

Candidate Triggers
current event +
stored state

Semantic Pattern
Anomaly Variant Match
Temporal Emotional
Associative

Intervention Controller

Selective Checks

Phase Gate
Variant Match

Think (optional)
Pattern
Anomaly
ambiguous
Variant Match

Candidate Score

α relevance
 β urgency
 γ confidence
 δ novelty
select highest

Decision Gates

Threshold
score $> \theta_{\text{push}}$

Per-habit
Cooldown
trigger + source
+ habit key

EVALUATED

OPTIONAL

Structured
PUSH

Natural-language
realization
content only

Ambient
Memory
injection

Observed
agent outcome

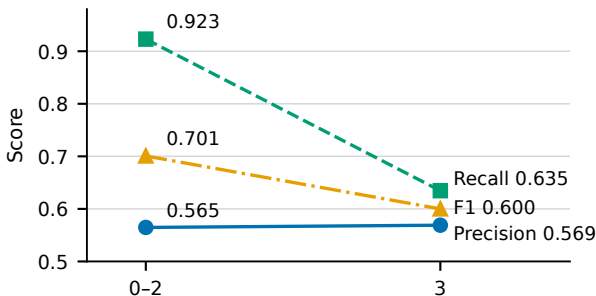
STRUCTURED NO-PUSH / SUPPRESS

no trigger | feedback cooldown | phase/Think suppression
below threshold | per-habit cooldown

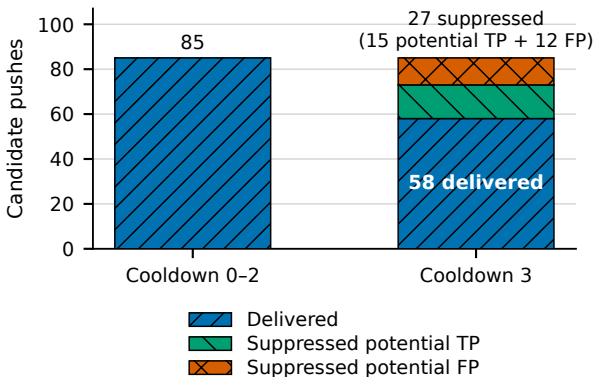
Decision trace: trigger hits, candidates and scores, selected candidate,
phase/Think/cooldown state, outcome, and suppression reason.

Outcome feedback updates Habit Store and global cooldown

(a) Shopping metrics at tested points



(b) Delivered and suppressed pushes



Only tested points are shown; the connector is a visual guide.